

Kelly Xu

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EDUCATION

Duke University, Durham, NC

Expected May 2026

GPA: 3.90, B.S. in Computer Science, Concentration in Software Systems

Involvement: Director of Duke Applied Machine Learning, Developer for Duke Christensen Center of Innovation

Coursework: Software Engineering, Operating Systems, Database Systems, Computer Architecture, Computer Network Architecture, Web Development, Design and Analysis of Algorithms, Algorithmic Machine Learning

SKILLS

Languages: JavaScript, TypeScript, Python, Java, SQL, HTML

Frameworks and Dev Tools: AWS, React, React Native, Flask, NodeJS, ExpressJS, Docker, PostgreSQL, Postman

EXPERIENCE

Software Engineer Intern | Carters Inc

May 2025-present

- Modernizing legacy AWS infrastructure and CI/CD pipelines for Carter's loyalty platform, enhancing system maintainability and reliability at scale.
- Developed and deployed two production-grade APIs that improved product recommendations and boosted customer engagement for millions of users.

UI/UX Design Intern | Merck, Rahway NJ

Jun 2024 - Aug 2024

- Redesigned and rebuilt a critical internal website, improving usability for 1,000+ users.
- Developed an automation tool for website content migration, reducing migration time by 90% and minimizing data loss, adopted by the Merck IT department.
- Analyzed 7.5 years of website traffic data, uncovering key user behavior trends.

Software Engineer Intern | Duke Office of Information Technology, Durham NC

May 2023 - Aug 2023

- Developed a server activity logging application for phishing attack prevention using Python and socket programming, deployed on Duke's network in 4 weeks.
- Implemented client configurable settings and Docker containerization for integration with the STINGAR cybersecurity platform, in collaboration with Forewarned, Inc.

Teaching Assistant: Data Structures and Algorithms | Duke, Durham NC

Aug 2024 - present

- Teaching weekly classes of around 30 students, improving understanding and engagement through effective communication of data structures concepts and algorithms analysis in Java.

PROJECTS

Irex | Duke Database Research Group, Durham NC

Jan 2025 - present

- Developing the frontend for Irex, an educational SQL debugger used by hundreds of students.

Amazon Plant Marketplace

Sept 2024 - Dec 2024

React, Axios, JavaScript, HTML, CSS, Python/Flask, PostgreSQL, Figma

- Designed the PostgreSQL database and developed a RESTful Web API using Flask for an online plant marketplace with features for browsing, buying, selling, and reviewing products.
- Implemented an optimized product search and filtering system for 9,000+ plant products, improving performance through efficient database queries and pagination.

Duke Social Groups Event Manager

Sept 2024 - Apr 2025

React Native, Expo, TypeScript, JWT, HTML, CSS, NodeJS, ExpressJS, Prisma ORM, PostgreSQL

- Building a cross-platform event management solution for web and iOS in a team of 7.
- Led the design and development of the user profiles feature, integrating token-based authentication and enabling users to view and manage past and upcoming events.